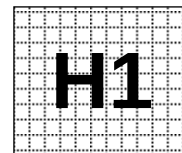


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H1 BIOLOGY

8875/2

Paper 2: Core (Mark Scheme)

Friday

16 September 2011

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/6
3	/11
4	/12
5 or 6	/20
Total	/60

This document consists of **11** printed pages.

[Turn over

Section A

Answer **all** questions.

1.

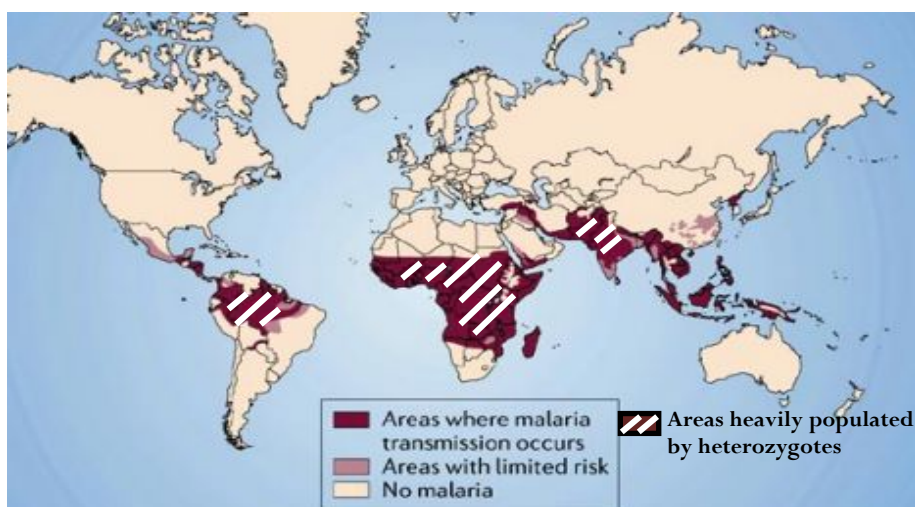
(a) Haemoglobin is an efficient oxygen transport molecule in red blood cells. With reference to its tertiary and quaternary structures, explain how haemoglobin is adapted to its function.

.....[3]

- 1 globular structure;
- 2 allows many haemoglobin molecules to be packed into a red blood cell;
- 3 (tertiary structure) consists of polypeptide chain folding with majority of its hydrophobic amino acid residues buried in the interior and the majority of its hydrophilic amino acid residues on the outside;
- 4 enables haemoglobin to be soluble in an aqueous medium;
- 5 quaternary structure as it is made up of 2 α -chain subunits (141 amino acids) and 2 β -chain subunits (146 amino acids);
- 6 each subunit can bind one oxygen molecule to increase number of oxygen molecules carried;
- 7 haem prosthetic group associated with each subunit;
- 8 enables the reversible binding of oxygen by the Fe^{2+} ion;

(b) Sickle cell anaemia is a recessive disease whereby the normal haemoglobin (encoded by Hb^A allele) in red blood cells is replaced entirely by abnormal haemoglobin (encoded by Hb^S allele). Abnormal haemoglobin crystallises when oxygen level is low and the crystals causes the red blood cells, which are typically biconcave, to change to a sickle shape. Such sickle shaped red blood cells have a lower lifespan.

Fig. 1.1 shows the areas where malaria is prevalent. Malaria is a disease caused by a *Plasmodium* parasite which spends part of its life cycle inside the red blood cells.



Adapted from <http://www.dailymail.co.uk/travel/article-593842/How-avoid-malaria.html>

Fig 1.1

(i) Based on Figure 1.1 and your knowledge on relationship between alleles, suggest explanations for the distribution of heterozygotes.

.....[2]

1 heterozygotes are found in large numbers in malaria-infested areas;

2 alleles Hb^A and Hb^S are co-dominant;

3 heterozygotes produce both normal and abnormal haemoglobin;

4 red blood cells containing abnormal haemoglobin die easily, trapping the parasites and stopping infection

/ slow-down in blood flow in heterozygotes hampered the parasite's ability to travel and infect new cells;

5 heterozygotes have higher chance of survival against malaria/ at selective advantage in malaria-infested areas;

Asita and Ayoo is an African couple who are free from symptoms of sickle-cell anemia. Both their parents appear normal but they have siblings who died young from the disease. Asita has blood type O while Ayoo has blood type AB.

Provided that their first child is diseased, use a genetic diagram to calculate the probability that their second child will be a boy with blood type B and normal haemoglobin. State clearly, the types of haemoglobin present in each phenotype.

.....[4]

Parental phenotypes:	Blood O; Normal and abnormal Haemoglobin	x	Blood AB; Normal and abnormal Haemoglobin
Parental genotype:	$I^O I^O Hb^A Hb^S$		$I^A I^B Hb^A Hb^S$
Parental gametes:	$I^O Hb^A$ $I^O Hb^S$		$I^A Hb^A$ $I^A Hb^S$ $I^B Hb^A$ $I^B Hb^S$

Punette square showing F_1 genotypes

	$I^A Hb^A$	$I^A Hb^S$	$I^B Hb^A$	$I^B Hb^S$
$I^O Hb^A$	$I^A I^O Hb^A Hb^A$ Blood A; normal haemoglobin	$I^A I^O Hb^A Hb^S$ Blood A; normal and abnormal haemoglobin	$I^B I^O Hb^A Hb^A$ Blood B; normal haemoglobin	$I^B I^O Hb^A Hb^S$ Blood B; normal and abnormal haemoglobin
$I^O Hb^S$	$I^A I^O Hb^A Hb^S$ Blood A; normal and abnormal haemoglobin	$I^A I^O Hb^S Hb^S$ Blood A; abnormal haemoglobin	$I^B I^O Hb^A Hb^S$ Blood B; normal and abnormal haemoglobin	$I^B I^O Hb^S Hb^S$ Blood B; abnormal haemoglobin

F₁ phenotype: Blood A, normal : Blood A, both : Blood A, abnormal: Blood B, normal: Blood B, both: Blood B, abnormal
 F₁ phenotypic ratio: 1 : 2 : 1 : 1 : 2 : 1

Probability that second child is a boy with blood type B and normal haemoglobin = $\frac{1}{2}$
 $\times \frac{1}{8} = \frac{1}{16}$

- 1 parental genotypes – I⁰I⁰Hb^AHb^S;
- 2 parental genotypes – I^AI^BHb^AHb^S;
- 3 parental gametes – I⁰Hb^A, I⁰Hb^S; I^AHb^A, I^AHb^S, I^BHb^A, I^BHb^S;
- 4 F₁ genotypes ;
- 5 Punette square with legend/ F₁ genotypes corresponds to phenotypes ;
- 6 F₁ phenotypic ratio;
- 7 correct presentation of genetic diagram;
- 8 correct calculation ;

(ii) Account for the difference in amino acid sequence between normal and abnormal haemoglobin.

.....[2]

1 glutamic acid at position 6 is being substituted by valine in abnormal haemoglobin;

2 mutation at gene level;
 3 ref to base substitution;

4 thymine (T) at the 17th nucleotide of the gene is replaced by adenine (A);

[Qn1 - Total: 11]

2.

(a) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This drug slows down cell division by acting on one of the enzymes dihydrofolate reductase, in a metabolic pathway.

This enzyme catalyses the conversion of dihydrofolic acid to tetrahydrofolic acid, that results in the formation of purines. The chemical structures of both dihydrofolic acid and Methotrexate are shown below in Fig. 2.1.

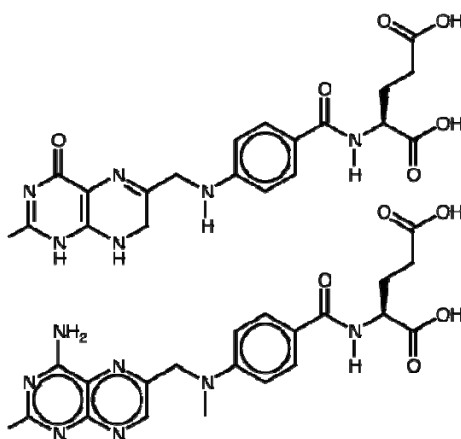


Fig 2.1
Chemical structures of
dihydrofolic acid (top)
methotrexate
(bottom).

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

.....[3]

- 1 Methotrexate is a competitive inhibitor;
- 2 as it has a structural resemblance to dihydrofolic acid;
- 3 Methotrexate competes with the substrate;
- 4 for binding at the active site of the enzyme;
- 5 Inhibition leads to less purines formed;
- 6 and therefore less nucleotides synthesized leading to less DNA replication;

(b) Describe briefly how the information on DNA is used to synthesize the RNA transcript for dihydrofolate reductase.

.....[3]

- 1 General transcription factors bind;
- 2 to promoter sequence / TATA box;
- 3 RNA polymerase II binds to promoter, (DNA helix unzips);
- 4 Adds complementary ribonucleotides/base paired ribonucleotides;
- 5 to the 3' end of growing RNA chain/ new strand made in 5' to 3' direction ;
- 6 Termination of transcription controlled by polyadenylation signal;

[Qn2 - Total: 6]

3.

Fig. 3.1 shows the diagram of a mitochondrion and a chloroplast.

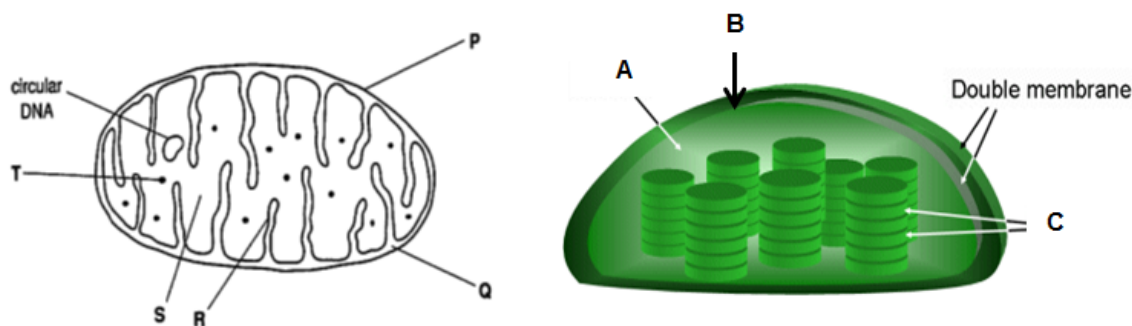


Fig. 3.1

(a) In each case, state the letter(s) that indicate(s) the site of: [2]

(i) Carbon fixation (**A**)

(ii) ATP synthesis (**S, R, C**)

(b) Suggest one function for the circular DNA in mitochondria as shown in Fig. 3.1.

.....[1]

- 1 DNA template for synthesis of proteins involved in respiration;
- 2 e.g. enzymes/ ATP synthase/ electron carriers of the Electron Transport Chain;

(c) Briefly outline the process of ATP production in oxidative phosphorylation.

.....[4]

- 1 electrons are transferred from reduced NAD (NADH) and FADH₂;
- 2 down a series of electron carriers/ ETC (of progressively lower energy levels);
- 3 energy released during electron transfer;
- 4 is used to pump H⁺ from the mitochondrial matrix into the intermembrane space;
/ H⁺ concentration gradient is established;
- 5 H⁺ diffuses back into the mitochondrial matrix;
- 6 through stalk particles;
- 7 ATP is synthesized;
- 8 from ADP and inorganic phosphate;;

(d) ATP exists as charged molecules in cells. Suggest why ATP made in the light reaction of photosynthesis can only be used for the dark reaction in the chloroplast and not for other cellular processes.

.....[1]

- 1 No protein transporters present to transport (charged) ATP;
- 2 across the hydrophobic core of the phospholipid bilayer of chloroplast membranes;

(e) Fig.3.2 shows the level of production of the products of photophosphorylation in a plant.

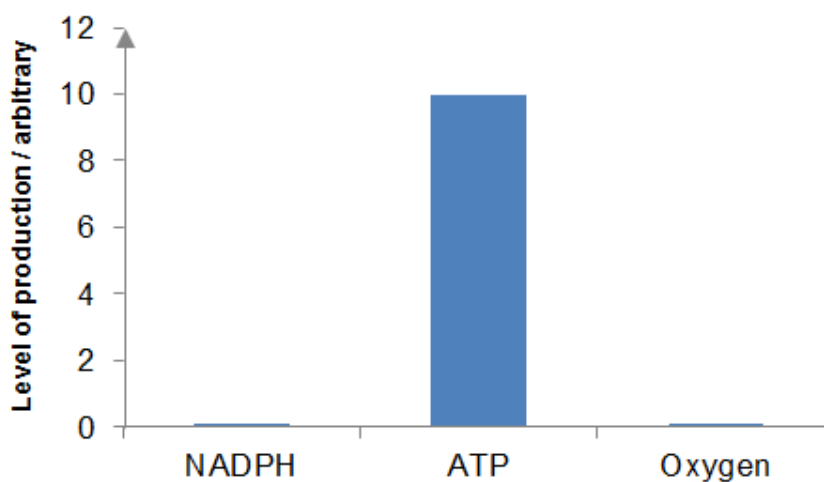


Fig. 3.2

With reference to Fig. 3.2, account for the level of synthesis of NADPH, ATP and oxygen.

.....[3]

- 1 negligible level of production of NADPH and oxygen produced;;
- 2 high level of production of ATP;
- 3 as shown by 10 arbitrary units for ATP;
- 4 plant is undergoing cyclic mode of photophosphorylation/ electrons at PS I are transferred to the ETC between PS II and PS I before going back to PSI;
- 5 allowing ATP synthesis by chemiosmosis;
- 6 No reduction of NADP⁺ as electrons are not transferred to ETC after PS I/final electron acceptor is not NADP⁺;
- 7 no oxygen released due to no photolysis/ PS II not involved;

[Qn3 - Total: 11]

4

A molecular scientist is in the process of cloning the human anti-thrombin gene using bacteria cells.

Fig.4.1 shows 2 plasmids (A and B) and the respective restriction sites of various restriction enzymes. In plasmid B, mutations were deliberately introduced and affected restriction sites are denoted by *. The lengths of the plasmid DNA between consecutive restriction sites are also indicated.

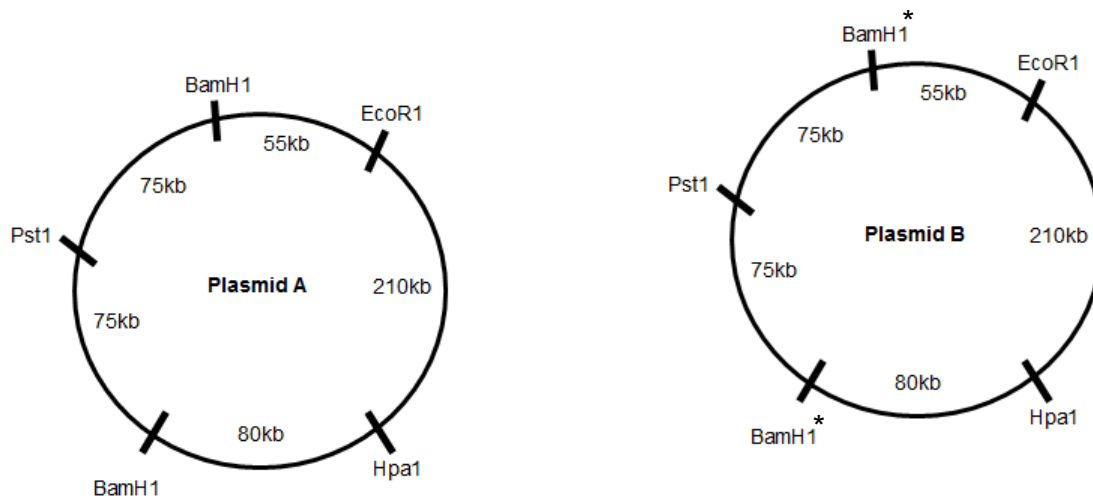


Fig. 4.1

He carried out a series of 2 experiments by adding different restriction enzymes to the plasmids. The components added to each tube can be seen in **Table 4.1**.

Each tube was then left to incubate at 37°C for about 15 minutes (sufficient time for complete digestion to take place).

Tube number	Tube Contents
1	Plasmid A + <i>BamH1</i> + <i>Hpa1</i>
2	Plasmid B + <i>BamH1</i> + <i>EcoR1</i> + <i>Pst1</i>

Table 4.1

(i) Describe the natural role of *BamH1*, *EcoR1*, *Pst1* and *Hpa1* in the bacteria in which they are produced.

.....[2]

- 1 to cleave;
 - 2 foreign DNA that enters the bacterial cell;
 - 3 protect bacterial cells;
 - 4 from bacteriophage/viruses;
- (R) cut at, restriction site/gene; used in GE**

(ii) The DNA fragments obtained were then separated by gel electrophoresis. A DNA ladder was loaded into the left most well of the agarose gel. The biggest fragment of the ladder is 550kb and the difference between each fragment of the ladder is 50kb.

The molecular biologist then loaded the mixture from each tube into their respectively numbered wells of the agarose gel as seen in Fig. 4.2.

Predict the results of the gel electrophoresis by drawing in the bands for each of the 2 tubes in Fig. 4.2 below. Indicate the length of each band. [2]

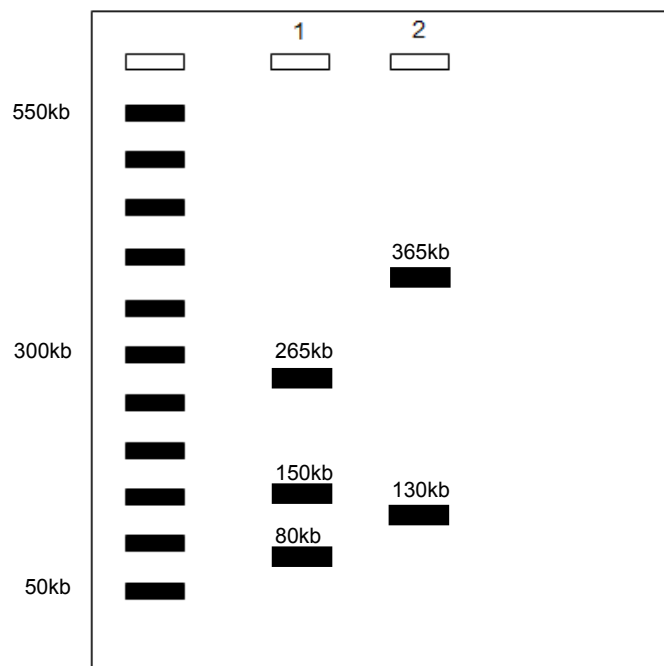


Fig. 4.2.

- 1 correct position of all bands in tube 1;
- 2 correct position of all bands in tube 2;
- 3 correct length indicated of all bands in tube 1 and 2;;

A genetic engineer cuts plasmid A shown in Fig. 4.1. with *Hpa*I in order to insert a human anti-thrombin gene.

(iii) Outline the steps that must subsequently be followed to produce a recombinant plasmid.

.....[2]

- 1 cut the human gene using the same restriction enzyme / *Hpa*I;;
- 2 add ligase;
- 3 to seal the nicks in the sugar-phosphate backbone / form phosphodiester bonds between the ends of the cut plasmid and human gene;

Fig. 4.3 shows the results of plating the transformed bacteria cells onto an agar plate with the appropriate substances.

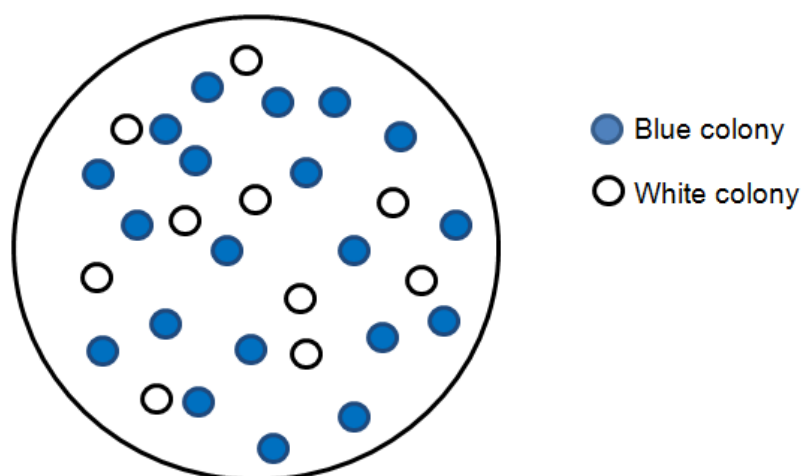


Fig. 4.3

(iv) Calculate the percentage of cells into which the recombinant plasmid had been successfully introduced.

.....[1]

1 percentage of cells containing recombinant plasmid = $10/30 \times 100 = \underline{33.3\%}$;;

(v) Explain why some colonies appeared white while others appeared blue.

.....[3]

- 1 agar plate contained X-gal and IPTG (so that cells that contain recombinant plasmids can be easily identified);;
- 2 blue colonies contain the reannealed / non-recombinant plasmids;
- 3 that contain intact lacZ gene;
- 4 white colonies contain recombinant plasmids;
- 5 with lacZ gene in the plasmid disrupted due to insertion of foreign DNA / gene of interest;
- 6 ref. to lac z gene coding for β -galactosidase enzyme;
- 7 which catalyses the conversion of X-gal in the agar from colourless to blue;

(vi) The same molecular scientist has problems obtaining functional anti-thrombin protein in this cloning experiment. Analysis reviewed a much longer amino acid sequence in the non-functional protein. Suggest and explain the reason why.

.....[2]

- 1 DNA/gene containing introns was used to insert into the plasmid;;
- 2 bacteria cells lack RNA splicing machinery;
- 3 thus introns are being expressed into amino acids/gene is not expressed correctly;

[Qn4 - Total: 12]

Section B

Answer **EITHER 5 OR 6** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

- 5** **(a)** Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell. [8]
- (b)** Describe the functions of triglycerides and phospholipids in living organisms. [4]
- (c)** Describe the main features of collagen that contribute to its tensile strength [8]

[Total: 20]

Or

- 6** **(a)** Describe the development of cancer as a multi-step process. [6]
- (b)** Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells and explain their normal functions in a living organism. [8]
- (c)** Explain how natural selection may bring about evolution. [6]

[Total: 20]

5(a) Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell.

[8]

Centrioles

- 1 Rod-like structures ;
- 2 Positioned at right angles to each other ;
- 3 Exists as a pair ;
- 4 Consists of 9 triplets of microtubules;
- 5 Fn : organization of spindle fibres during nuclear division;
- 6 Important for separation of chromosomes / chromatids during meiosis /mitosis

Nucleolus

- 7 Densely stained granules and fibres ;
- 8 Observable in a non-dividing cell ;
- 9 One of more nucleoli may be present within nucleoplasm ;
- 10 Fn : site of transcription of rRNA genes / synthesis of rRNA ;
- 11 Important for formation of ribosomal subunits involved in translation / protein synthesis
/ component of ribosomes

Smooth endoplasmic reticulum

- 12 Consists of a network of fine tubules;
- 13 Appears smooth due to the lack of ribosomes ;
- 14 Fn : synthesis of lipids ;
- 15 Required by the cell as structural component of membrane e.g. phospholipids
/ Required by the cell as components of cell-cell recognition or cell-cell adhesion
e.g. glycolipids
- 16 Fn : detoxification of drugs and poisons ;
- 17 Prevents built-up of toxic materials which may harm the cell ;
- 18 Fn : storage of Calcium ions (in muscle cells) ;
- 19 Required for muscle contraction as Ca^{2+} functions as a second messenger ;

5(b) Contrast triglycerides and phospholipids.

[4]

	Triglycerides	Phospholipids
Structure	Consist of <u>three</u> fatty acids covalently bonded to a glycerol molecule ; No phosphate groups attached ; Or Do not have other groups attached to glycerol besides fatty acids ;	Consist of <u>two</u> fatty acids covalently bonded to a glycerol molecule ; Phosphate group attached to glycerol ; Or May have other groups attached to the phosphate group ; e.g. nitrogen-containing base choline
Properties	Hydrophobic / insoluble in water due to presence of hydrocarbon chains ;	Amphipathic due to presence of both hydrophilic and hydrophobic groups ;
Functions	Used for storage due to the lack of osmotic effect ; Used as a respiratory substrate due to its high energy per unit mass content compared to other substrates ; Required for the solubility of some fat-soluble vitamins due to their hydrophobic nature ;	Structural component of membranes in cells ; Involved in compartmentalisation for specialized conditions to be maintained in cells e.g. proton gradient in chloroplast, mitochondria

5(c) Relate the structural organization of collagen to its function.**[8]**

- 1 Each polypeptide chain is 1,000 amino acid residues long ;
- 2 Collagen is rich in hydrophobic amino acids ;
- 3 Ref glycine and proline ;
- 4 Collagen is insoluble in water
- 5 Due to the large molecular size of the tropocollagen molecule
- 6 helical conformation
- 7 triple helix / 3 polypeptide chains
- 8 called tropocollagen

Intramolecular bonds (between the 3 polypeptides)

- 9 cross-linked to each other ;
- 10 hydrogen bonds ;
- 11 between the –NH groups (of glycine residues) on each strand and –CO groups on the other two strands ;
- 12 small glycine residue fits into the crowded centre of the triple helix ;
- 13 presence of hydroxyproline also allows H bonds

Intermolecular bonds (between tropocollagen molecules)

- 14 intermolecular bonds between tropocollagen molecules stabilize these structures
- 15 Formation of covalent cross-links;
- 16 Involving lysine and hydroxylysine; allow tropocollagen molecules assemble to form bundles ;
- 17 Association into collagen fibrils ;
- 18 Aggregation into collagen fibres ;
- 19 Staggered parallel arrangement;
- 20 Allows collagen to provide great mechanical and tensile strength
- 21 Resists tears and stretching
- 22 Can function as component of connective tissues, cartilage and skin

6(a) Describe the development of cancer as a multi-step process.**[6]**

- 1 loss of function mutation of tumour suppressor gene
- 2 mutant tumour suppressor alleles are usually recessive
/ mutations must knock out both alleles in a cell's genome to produce abnormal protein
- 3 loss of arrest of cell division to allow more time for the cell to repair its DNA / loss of ability for DNA repair / loss of apoptosis
- 4 gain of function mutation of proto-oncogene
- 5 oncogenes behave as dominant alleles
/ only need one mutated allele to produce abnormal protein
- 6 lead to stimulation of the cell cycle / cell keeps dividing
- 7 accumulation of many mutations
- 8 activation of telomerase gene
- 9 telomerase enzyme prevents the shortening of the chromosome ends / cell can continue to divide indefinitely
- 10 angiogenesis / formation of new network of blood vessels to the cancer cells
- 11 blood vessels provide the cancer cells oxygen and nutrients for growth and to remove any waste products
- 12 loss of contact inhibition / density-dependence (and cells do not stop dividing)
- 13 loss of ability to differentiate
- 14 formation of benign tumour
- 15 cells in the tumour no longer exhibit anchorage dependence / loss of cell adhesion
- 16 metastasis / break loose and may enter the bloodstream / invade other tissues form secondary tumors
- 17 tumour is considered malignant

6(b) Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells and explain their normal functions in a living organism. [8]

Zygotic stem cells

- 1 totipotent
- 2 cells in the morula (first few divisions after fertilization / fusion of an egg and sperm)
- 3 able to differentiate into any cell type / all the tissues of the body
- 4 including extra-embryonic tissues
- 5 daughter cells are pluripotent and multipotent
- 6 Function = give rise to the whole organism

Embryonic stem cells

- 7 pluripotent
- 8 inner cell mass of the blastocyst
- 9 have the ability to differentiate into almost any cell type to form any organ or type of cell
- 10 except the extra-embryonic tissues / placenta
- 11 daughter are not totipotent but are multipotent
- 12 Function = give rise to various organs in organism / multiple specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus

Blood stem cells

- 13 multipotent
- 14 able to differentiate into a limited range of cell type
/ produce only cells of a specific lineage
- 15 daughter cells are not pluripotent or totipotent
- 16 can differentiate into RBC, WBC, platelets
- 17 Function = replacement of worn out blood cells
/ daily blood turnover and fighting infections
- 18 lost through normal wear and tear, disease, injury

6(c) Explain how natural selection may bring about evolution.**[6]**

- 1 populations have great reproductive potential
- 2 but numbers remain approximately constant
- 3 presence of variations in the form of different alleles
- 4 resulting from mutations

- 5 as many fail to survive because of competition for limited resources in response to selection pressures (eg. food, shelter, mates) / predation / diseases

- 6 individuals with genetic variations best adapted to the environment more likely to survive/ at selective advantage
- 7 reproductive age and pass on their beneficial genes to offspring
- 8 individuals with genetic variations not well adapted to the environment less likely to survive/ at selective disadvantage
- 9 such individuals are less likely to survive and reproduce to pass down the alleles to their offspring

- 10 selection of advantageous phenotypes lead to change in frequency of particular alleles and genotypes
/ proportion of individuals possessing the advantageous characteristic increases whereas that lacking the characteristic decreases

- 11 lead to evolutionary change over a long period of time
- 12 new species arising from existing species / ref. speciation theory